

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Web Development”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

**BACHELOR OF ENGINEERING IN Department of
Artificial Intelligence and Machine Learning**

Submitted by:

ASHISH GANGWAR – 1DS20AI013

Conducted at



Zapcom.com

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

2023-24

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

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CERTIFICATE

This is to certify that the Internship titled “**Project Trainee**” carried out by **ASHISH GANGWAR[1DS20AI013]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head
Poornima Hiregoudar
Tech Lead
Zapcom

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **ASHISH GANGWAR**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Zapcom**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date:

USN: 1DS20AI013

Place: Bangalore

NAME: ASHISH GANGWAR

OFFER LETTER



21st November 2023

Mr. Ashish Gangwar
Collage ID: 1DS20AI013

LETTER OF INTERNSHIP

Dear Ashish,

Congratulations!

Consequent to your conversation held with us, we are pleased to offer you an internship as "Intern" at Zapcom Solutions Pvt Ltd on the terms and conditions mentioned below:

Current Place of Posting: Bangalore

Stipend: Rs.25,000/- per month

Internship Start Date: 12th February 2024

This internship is for a period of **6 months** and based on your performance the company will evaluate full time opportunity for you with the organization. You are obliged to observe the work regulations in force at Zapcom, as far as the punctual beginning and ending of work within the working hours binding for the employees of Zapcom, the discipline at work and the reliable work performances are concerned.

During that term, company may terminate this agreement for any reason or no reason with fifteen (15) days' notice. The terms of this Agreement shall be governed by and interpreted in accordance with the laws of India.

You will be given a laptop and its accessories during the training in Zapcom. The same will have to be returned by you to Zapcom on completion of the training.

Please sign and return the duplicate copy of this letter in token of your acceptance. We look forward to welcoming you in the organization on Monday, **12th February 2024** at 11 am.

Yours faithfully,

For **ZapCom Solutions Pvt Ltd,**

Accepted

SRINIVAS KOTHAKOTA
COO

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COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr Poornima Hiregoudar, Zapcom, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ASHISH GANGWAR
[1DS20AI013]

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CHAPTER 1

INTRODUCTION

At Zapcom, innovation is woven into the fabric of our culture, driving us to constantly explore new frontiers in enhancing the digital experience for our clients and their customers. Recently, we embarked on an exhilarating journey to integrate a state-of-the-art chatbot into our website, marking a significant leap forward in how we engage with visitors in real-time. Powered by React.js, a renowned JavaScript library celebrated for its flexibility and performance, our chatbot seamlessly integrates into our website, offering an intuitive and dynamic platform for interaction.

Our collaboration with React.js transcended mere integration; it became the cornerstone of our quest to fashion a truly user-centric chatbot interface. With painstaking attention to detail, our team ensured that the chatbot not only enhances accessibility but also delivers unparalleled responsiveness across a myriad of devices and browsers. React.js provided the robust foundation necessary for our chatbot to adapt and evolve alongside the evolving needs of users, ensuring a seamless and personalized experience for every visitor.

However, our dedication to innovation extends far beyond the initial integration phase. By harnessing the power of React.js, we've established a sturdy framework for future enhancements and optimizations. This framework empowers us to continually refine and elevate the capabilities of our chatbot, enabling us to stay ahead of the curve in an ever-evolving digital landscape. At Zapcom, innovation is more than just a goal; it's a guiding principle ingrained in every facet of our operations.

In conclusion, our journey with React.js has not only revolutionized our approach to online engagement but has also opened up new vistas of possibilities. By leveraging React.js to power our chatbot integration, we're poised to redefine the standards of online interaction and set new benchmarks for excellence in the digital realm. At Zapcom, innovation is our compass, and with React.js leading the way, we're primed to chart a course toward a future where possibilities are limitless, and boundaries are mere stepping stones to greater achievements.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Zapcom stands as a premier global engineering and product development powerhouse, dedicated to serving an expansive array of industries with tailored software solutions crafted to meet the exacting demands and strategic objectives of their clients. Renowned for their prowess in crafting bespoke software, Zapcom's multifaceted services span a comprehensive spectrum, encompassing vital domains such as cutting-edge data management, cloud solutions engineered for peak performance, robust security frameworks, and sophisticated enterprise-level applications finely tuned to optimize operational efficiency.

Zapcom Pvt Solution Ltd stands as a beacon of excellence and innovation in the industry. Founded with a vision to redefine industry standards, Zapcom is committed to delivering outstanding solutions tailored to meet the diverse needs of its clients. The company's dedication to excellence is evident in every aspect of its operations, from the inception of ideas to their execution. Quality, reliability, and customer satisfaction are not just goals but ingrained standards that guide every action taken by the organization. Innovation lies at the heart of Zapcom's ethos, with a dedicated team continuously striving to innovate and adapt to the rapidly evolving landscape. By harnessing the latest technologies and methodologies, Zapcom develops cutting-edge solutions that address the evolving needs of its clients and the market.

Setting themselves apart with an unwavering commitment to innovation, Zapcom is driven by a steadfast resolve to future-proof their solutions, ensuring they remain not just relevant but indispensable in the ever-evolving landscape of technology and business. Their holistic approach extends beyond mere technological prowess; it is underpinned by a profound dedication to facilitating lasting transformation within the businesses they serve. By fostering a culture of positive change, Zapcom endeavors to empower their clients to thrive in dynamic environments, driving sustainable growth and resilience in the face of emerging challenges.

In essence, Zapcom's ethos is one of continuous evolution, where innovation, adaptability, and a genuine passion for driving meaningful change converge to redefine the boundaries of possibility in the realm of software engineering and product development. With a steadfast commitment to excellence and a vision firmly rooted in the future, Zapcom stands as a trusted partner poised to navigate the complexities of today while charting a course towards tomorrow's triumphs.

CHAPTER 3

SCOPE OF WORK

The React.js and Spring Boot training program is a comprehensive initiative designed to equip participants with both theoretical understanding and practical skills necessary for excelling in frontend and backend web development. Beginning with an in-depth exploration of React.js, participants learn about its fundamental principles, including component-based architecture, which serves as the foundation for creating reusable UI components. Advanced topics such as state management and routing are covered to enhance participants' abilities in building interactive web applications.

Additionally, the program delves into advanced React.js concepts like hooks, context API, and Redux, empowering participants to develop flexible and efficient frontend solutions. In parallel, participants gain proficiency in Spring Boot, learning to build scalable backend services using RESTful APIs and Spring Data JPA for database integration. Security measures, including authentication and authorization techniques like JWT, are also emphasized to ensure robust protection of sensitive data.

Practical experience is emphasized through hands-on exercises, interactive discussions, and real-world projects, enabling participants to develop the expertise needed to create robust, full-stack web applications. Throughout the training, experienced instructors and mentors provide dedicated support, guiding participants through the curriculum and fostering an environment conducive to learning and growth.

In conclusion, the React.js and Spring Boot training program prepares participants for success in the dynamic field of web development by providing a structured curriculum that combines theoretical knowledge with practical skills. With a focus on both frontend and backend technologies, the program equips participants with the tools and expertise needed to excel as full-stack web developers in today's ever-evolving landscape of technology.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Embarking on the journey to master React.js and Spring Boot presents an exciting opportunity for personal and professional advancement. As I delve into React.js, I eagerly anticipate immersing myself in its flexible and component-based approach to constructing user interfaces. From creating reusable components to managing state and handling user interactions, my aim is to develop a profound understanding of React's fundamental principles and optimal techniques. Simultaneously, my exploration of Spring Boot promises to reveal a multitude of opportunities in backend development. With Spring Boot's efficient approach to building production-ready applications, I look forward to exploring topics such as designing RESTful APIs, integrating databases using Spring Data JPA, and implementing robust security measures.

As I progress through this learning journey, I envision synthesizing my newfound knowledge and skills to embark on the development of a comprehensive web application. This project will serve as the culmination of my efforts, incorporating React.js for the frontend and Spring Boot for the backend. From conceptualization to deployment, I anticipate encountering and surmounting challenges, refining my problem-solving abilities, and deepening my understanding of both frameworks. Furthermore, I view this project not only as an opportunity to achieve organizational objectives but also as a catalyst for innovation and creativity.

Beyond meeting immediate objectives, my aspiration is to leverage this project as a catalyst for ongoing learning and exploration. By pushing the boundaries of my current knowledge and embracing the iterative nature of software development, I aim to foster a mindset of adaptability and resilience. Ultimately, I envision this endeavor not merely as the completion of a project, but as a transformative journey that ignites my passion for technology and propels me toward new frontiers in the dynamic realm of software engineering.

CHAPTER 5

CHALLENGES AND SOLUTION

CHALLENGES

Throughout the learning phase and project development, several challenges emerged, each presenting valuable opportunities for growth and learning. Initially, grappling with the complexities of React.js posed a formidable hurdle, particularly in understanding concepts like component lifecycle and state management. Additionally, navigating the intricacies of Spring Boot, including setting up the environment and configuring dependencies, proved to be a steep learning curve. As the project progressed, integrating the frontend and backend components presented its own set of challenges, requiring meticulous attention to detail and troubleshooting skills to ensure seamless communication between the two layers. Moreover, managing project scope and timelines amidst the iterative nature of software development demanded effective time management and prioritization skills. Despite these obstacles, each challenge provided invaluable insights and fostered a deeper understanding of the technologies involved, ultimately culminating in a successful completion of the project and significant personal and professional growth.

SOLUTION

- ▶ **Thorough Understanding:** Study documentation and tutorials for React.js and Spring Boot to grasp core concepts.
- ▶ **Hands-On Practice:** Engage in coding exercises and small projects to reinforce learning and gain practical experience.
- ▶ **Seek Mentorship:** Consult experienced developers or mentors for guidance and assistance in navigating challenges.
- ▶ **Debugging Skills:** Develop strong debugging techniques to identify and resolve issues effectively during project development.
- ▶ **Effective Time Management:** Prioritize tasks and allocate time efficiently to ensure progress and meet deadlines.
- ▶ **Continuous Learning:** Stay updated with the latest advancements in React.js and Spring Boot through ongoing learning and exploration.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Integrating a chatbot into Zapcom's website stands as a pivotal moment in my journey of mastering React.js and Spring Boot, further enriched by the incorporation of TypeScript. This achievement symbolizes the culmination of tireless dedication, unyielding curiosity, and an insatiable thirst for knowledge. Through this project, I've not only honed my technical skills but also garnered invaluable experience in real-world application development.

Mastering React.js empowered me to craft dynamic and responsive user interfaces that seamlessly enhance user interaction. Concurrently, delving into Spring Boot equipped me with the necessary tools to architect robust backend services, ensuring the seamless functioning of the chatbot integration. However, it was the integration of TypeScript that truly elevated the project. Its addition introduced an additional layer of type safety and code clarity, significantly enhancing the quality and maintainability of the entire endeavor.

Reflecting on this accomplishment fills me with a profound sense of pride and satisfaction. Knowing that my contributions have made a tangible impact on Zapcom's digital landscape is immensely gratifying. Furthermore, this achievement serves as a testament to the limitless possibilities that arise from dedicated learning and persistent effort.

Yet, this milestone is not the culmination of my journey; rather, it serves as a launching pad for continued growth and innovation. The dynamic nature of software development demands perpetual adaptation and exploration. As such, I am committed to embarking on further endeavors, pushing the boundaries of what I know, and embracing new challenges with unwavering enthusiasm.

In essence, the integration of the chatbot into Zapcom's website represents not only a professional achievement but also a personal triumph. It underscores the transformative power of relentless pursuit and serves as a testament to the boundless potential that lies within each of us. With this accomplishment as a foundation, I eagerly anticipate the next chapter of my journey, eager to embrace the opportunities that lie ahead in the ever-evolving landscape of software development.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship at Zapcom has been a transformative journey characterized by significant accomplishments and profound personal growth. Throughout my tenure with the company, I played a pivotal role in integrating a chatbot into their website, leveraging my burgeoning expertise in React.js, Spring Boot, and TypeScript. This project served as a dynamic platform for translating theoretical knowledge into practical solutions, refining my problem-solving skills, and deepening my understanding of software development principles in a real-world context.

A fundamental lesson gleaned from this experience is the paramount importance of adaptability and resilience in the face of challenges. Whether grappling with the intricacies of React.js or navigating the complexities of Spring Boot, each obstacle presented an opportunity for growth and learning. Collaborating closely with colleagues and receiving invaluable mentorship from seasoned professionals at Zapcom further enriched my internship experience, providing me with diverse insights and perspectives that broadened my horizons.

Moreover, the integration of TypeScript into the project underscored the critical significance of writing clean, maintainable code and adhering to industry best practices in software engineering. Its type safety features not only enhanced the robustness of the codebase but also facilitated smoother collaboration and communication among team members, ultimately contributing to the project's success.

Reflecting on my internship journey fills me with a deep sense of gratitude for the opportunity to have made tangible contributions to Zapcom's initiatives while simultaneously expanding my skill set and expertise. This experience has not only bolstered my confidence as a developer but also ignited a fervent passion for continuous learning and professional growth. As I embark on the next chapter of my career, I am excited to leverage the knowledge and experiences gained during this internship to pursue future endeavors in the dynamic and ever-evolving realm of software engineering with enthusiasm and determination.

CHAPTER 8

RECOMMENDATIONS

To truly stand out and make the most of your internship at Zapcom, it's essential to adopt a holistic approach that encompasses various aspects of professional development. Firstly, diversifying your project portfolio can offer invaluable exposure to different technologies, methodologies, and team dynamics. Actively seek out opportunities to contribute to a range of projects, allowing you to broaden your skill set and gain a deeper understanding of various aspects of software development.

In addition to project work, leveraging mentorship and networking opportunities can provide invaluable guidance and insights from seasoned professionals within the organization. Don't hesitate to reach out to mentors for advice, feedback, and career guidance. Similarly, actively participating in networking events, team meetings, and cross-departmental collaborations can help you build meaningful relationships with colleagues, opening doors to new learning opportunities and potential career paths.

Continuous learning should be a cornerstone of your internship experience. Stay abreast of the latest industry trends, emerging technologies, and best practices through workshops, webinars, and self-directed study. Consider pursuing additional certifications or courses to deepen your expertise in areas of interest, demonstrating your commitment to ongoing growth and development.

Effective time management is crucial for optimizing productivity and ensuring the timely completion of tasks. Prioritize your workload based on urgency and importance, break larger projects into manageable tasks, and utilize productivity tools and techniques to stay organized and focused. By managing your time effectively, you can maintain a balance between project work, learning initiatives, and personal development activities.

Strong communication skills are essential for success in any professional environment. Practice active listening, articulate your ideas clearly and confidently, and seek clarification when needed to ensure mutual understanding among team members. Effective communication fosters collaboration, reduces misunderstandings, and enhances overall team productivity.

Finally, taking initiative demonstrates your initiative, ambition, and commitment to the organization's success. Look for opportunities to contribute beyond your assigned tasks, whether it's suggesting process improvements, proposing new project ideas, or volunteering for leadership roles within the team. By demonstrating proactive engagement and a willingness to go above and beyond, you can showcase your potential for leadership and make a lasting impression during your internship at Zapcom.

CHAPTER 9

CONCLUSION

As my internship at Zapcom draws to a close, I find myself reflecting on the transformative journey I've experienced over the past few months. It's been a period filled with invaluable learning opportunities, enriching experiences, and meaningful connections. From the outset, I embarked on this internship with a thirst for knowledge and a drive to expand my skills in software development. Through the guidance of mentors and the collaborative spirit of my colleagues, I've delved deep into the intricacies of technologies like React.js, Spring Boot, and TypeScript, gaining not only technical proficiency but also a profound understanding of their real-world applications.

Moreover, the diverse array of projects I've had the chance to work on has broadened my perspective and equipped me with a versatile skill set that I know will serve me well in my future endeavors. Whether it was troubleshooting issues in code, collaborating with cross-functional teams, or presenting project updates to stakeholders, each task presented its own set of challenges and learning opportunities.

But beyond the technical aspects, my internship at Zapcom has also been a journey of personal growth. I've learned the importance of effective communication, both in articulating my ideas and actively listening to others. I've honed my time management skills, juggling multiple tasks and deadlines with ease. And perhaps most importantly, I've cultivated a mindset of curiosity and resilience, embracing setbacks as opportunities for growth and approaching each challenge with optimism and determination.

As I bid farewell to Zapcom and look ahead to the next chapter of my journey, I do so with a deep sense of gratitude for the experiences I've had and the people I've had the privilege to work alongside. The lessons I've learned and the connections I've made will undoubtedly shape my future endeavors, empowering me to navigate the complexities of the professional world with confidence and conviction. My internship at Zapcom has been nothing short of transformative, and I am excited to carry the lessons learned and memories made with me as I embark on the next phase of my career.

CHAPTER 10

REFERENCES

1. <https://zapcg.com/>
2. <https://zapcom.ai/>